

Luiss

Libera Università Internazionale
degli Studi Sociali Guido Carli

Getting Started

Introduction to Computer Programming

Management and Artificial Intelligence

LUISS



How and Where

How can we run Python code?

Two approaches:

1. Interactive shell:

```
Python 3.10.12 (main, Nov 20 2023, 15:14:05) [GCC 11.4.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> print("Hello, World!")
Hello, World!
>>> █
```

- workflow:
 - start the interactive Python shell
 - write one line of code in the prompt after the symbols >>>
 - push enter to execute it and get the results
 - repeat as much as needed
- **not very user-friendly**
- appropriate only for quick tests
- nothing is saved: you will lose your code when exiting the shell!

How can we run Python code? (cont'd)

Two approaches:

2. Write the code into a textual file and then execute it:



- workflow:
 - write the code into a textual file with extension `.py`
 - execute the file with the Python VM
- **This is what you want to do!**
- Still *iterative* approach: write the code, check if it works, refine the code, ...

The code can be very cryptic...
Can we add some notes around it?

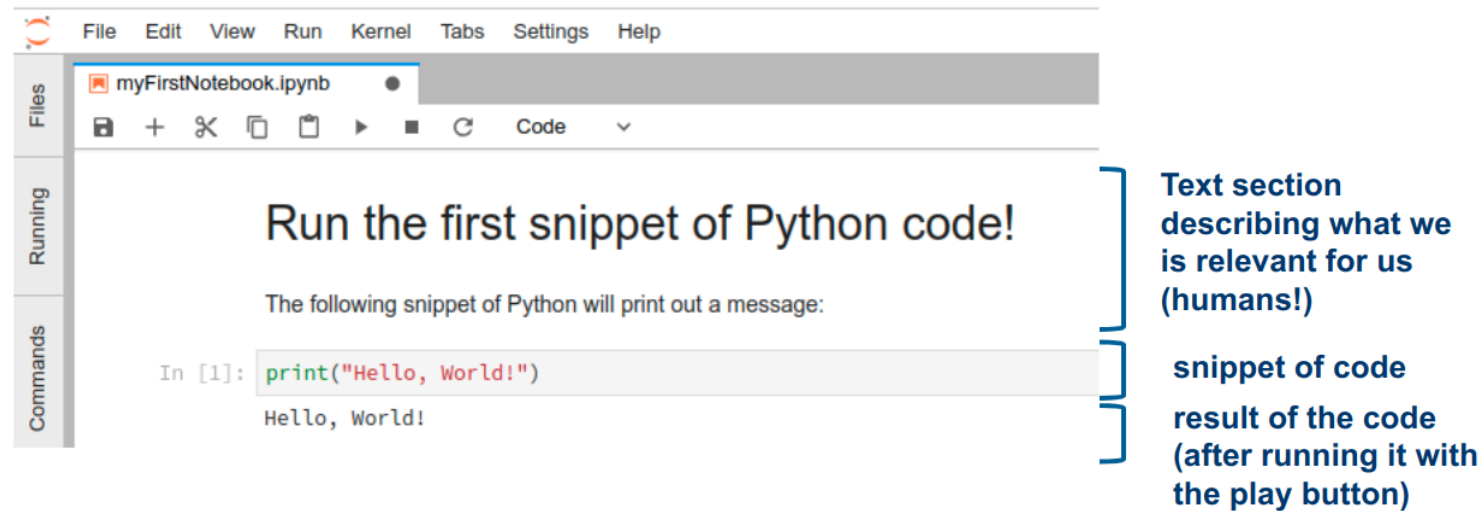
It could be hard to remember why we have written a piece of code, what are its effects and downsides.

It is often convenient to organize our code into "sections" and then add some text to describe what the code is expected to do. This is called:



Jupyter Notebook

Using the environment **JupyterLab** or with a proper Visual Studio Code setup, we can create a new notebook:



Where do we run our code/notebooks?

1. **Local setup:** your computer!



- you have to install Python and all the other bits
- limited by your computational power

2. **Cloud setup:**



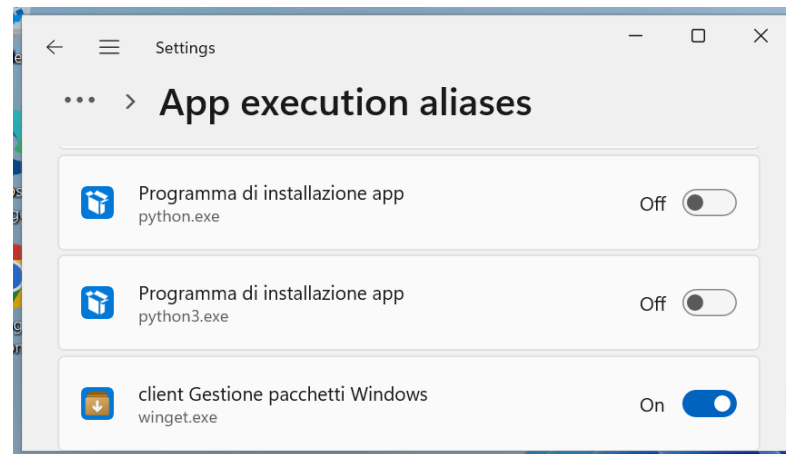
- no need to install everything
- limited by your money

Local Setup

[Only on Windows 11]

Disable Windows Store Aliases

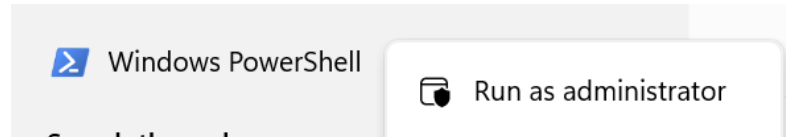
1. From the start menu, search and then open *Manage app execution aliases* (*Alias di esecuzione delle app*)
2. Disables aliases for *python.exe* and *python3.exe*:



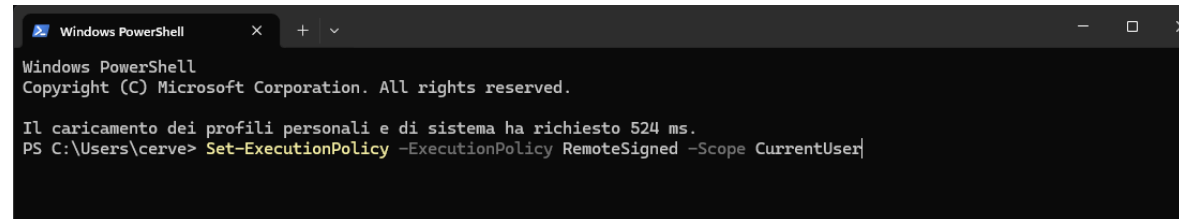
[Only on Windows 11]

Enable script cmdlet

1. From the start menu, search *Windows Powershell*, then using the right-click select *Run as administrator*.

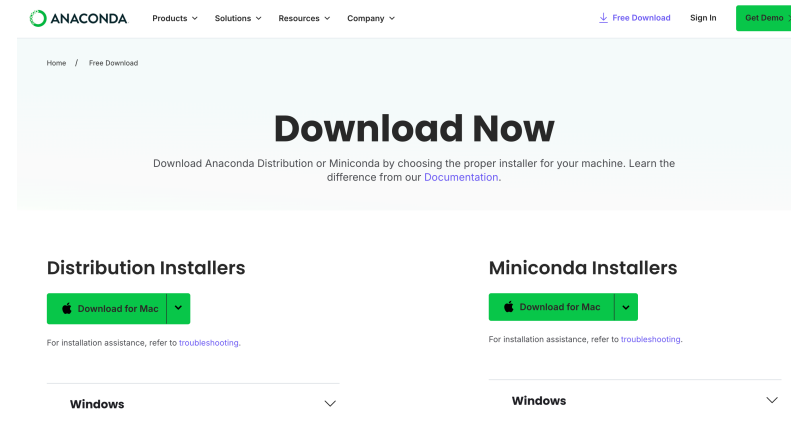
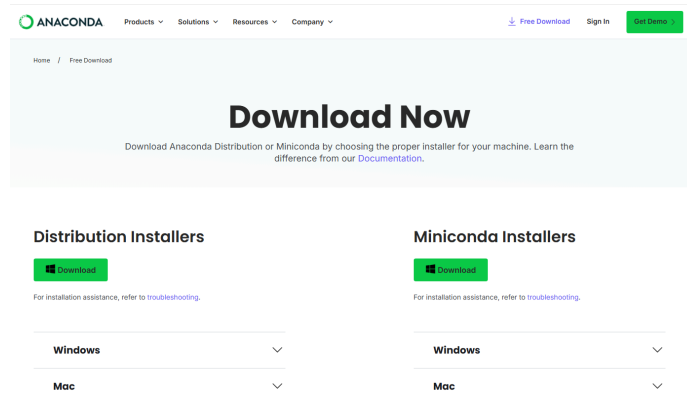


2. Type `Set-ExecutionPolicy -ExecutionPolicy RemoteSigned -Scope CurrentUser`, then run it:

A screenshot of a Windows PowerShell terminal window. The window title is 'Windows PowerShell'. The text inside the terminal shows the following: 'Windows PowerShell', 'Copyright (C) Microsoft Corporation. All rights reserved.', 'Il caricamento dei profili personali e di sistema ha richiesto 524 ms.', and the command prompt 'PS C:\Users\cerve> Set-ExecutionPolicy -ExecutionPolicy RemoteSigned -Scope CurrentUser'. The command is highlighted in yellow.

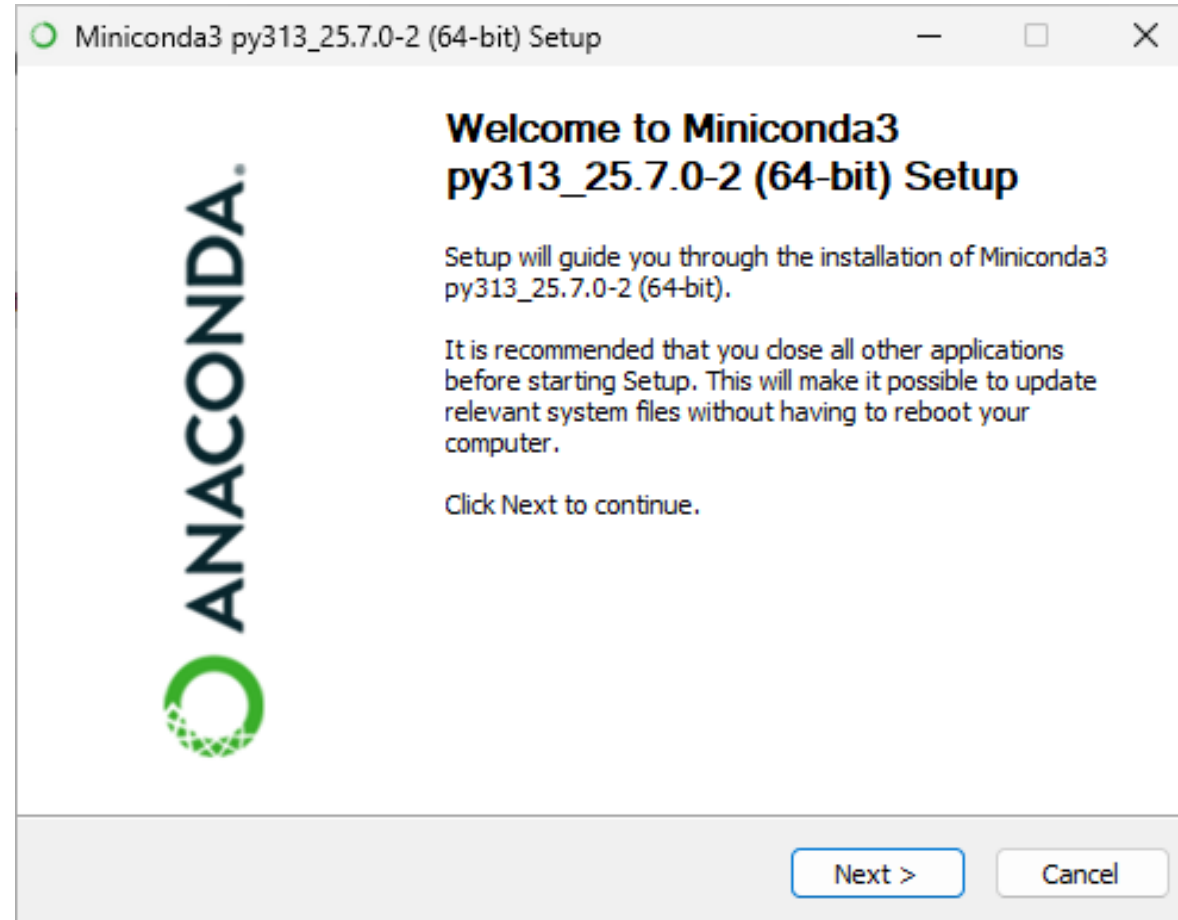
Installing Python via MiniConda

1. Open your web browser (e.g. Google Chrome, Microsoft Edge, Apple Safari, Mozilla Firefox) and go to <https://www.anaconda.com/download/success>
2. Click on the *Download* (for Windows) / *Download for mac* (for Mac) button under *Miniconda Installers* to get the installer



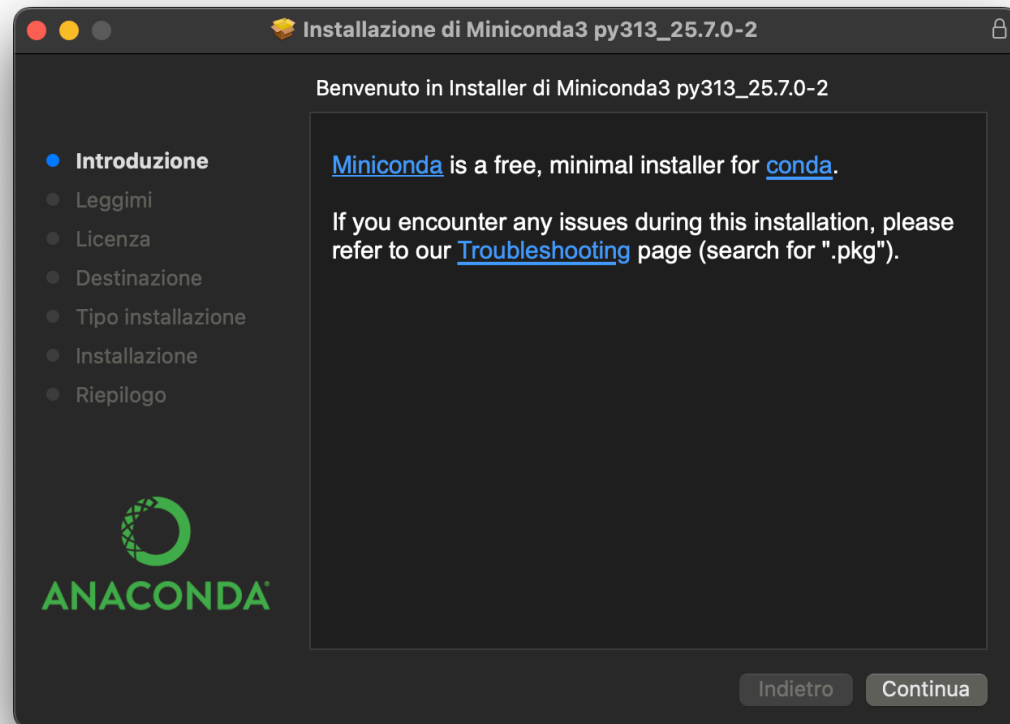
Launch the installer

Double click on the downloaded file (e.g. on Windows) to start the setup:



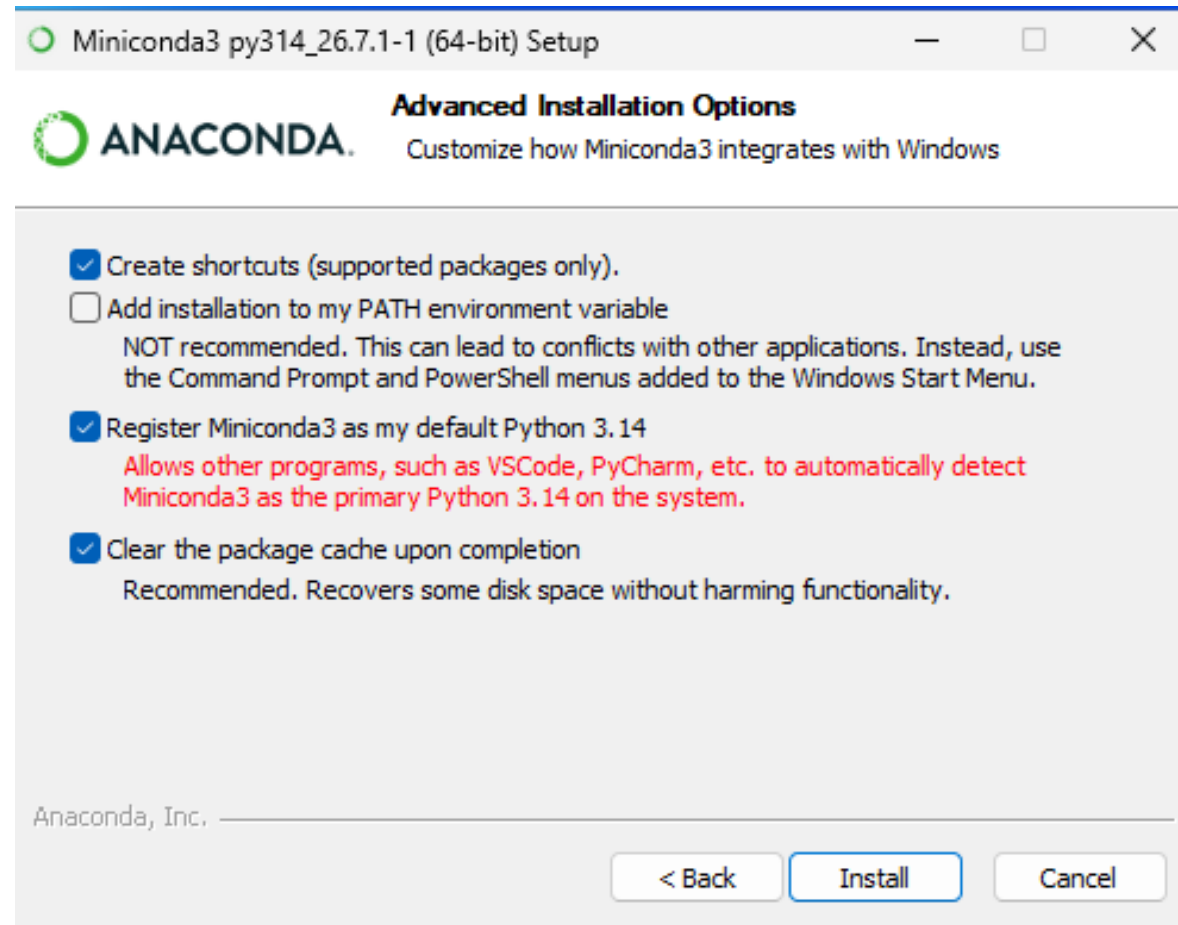
Launch the installer (cont'd)

Similarly on Mac OS X:



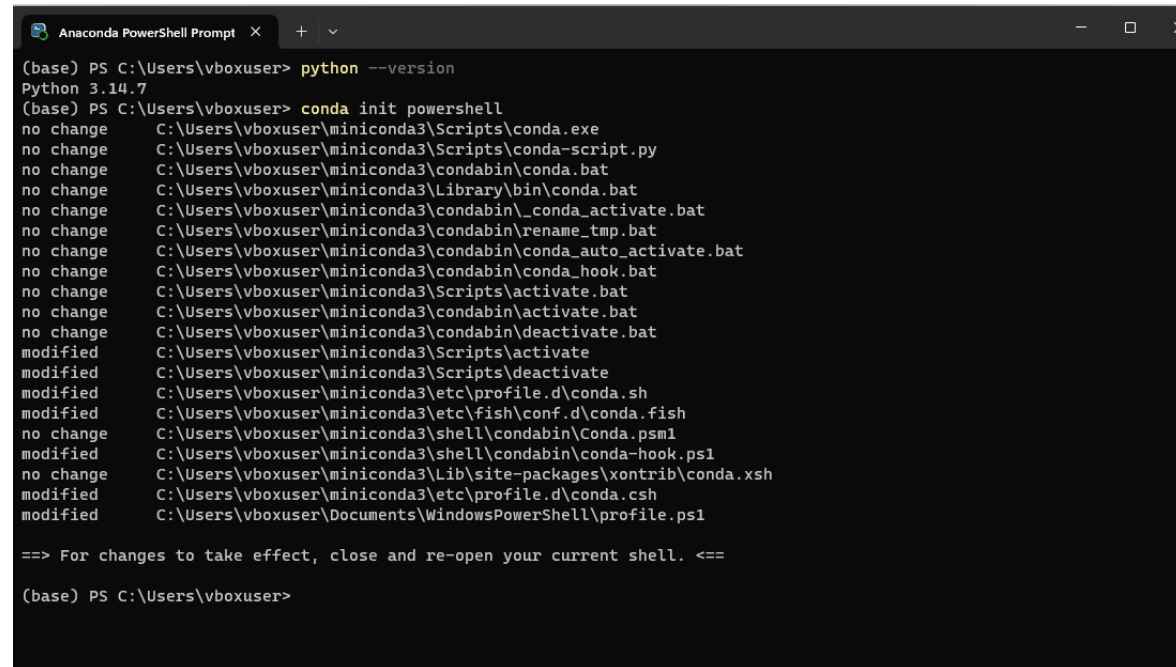
Launch the installer (cont'd 2) [Windows only]

- Install Miniconda just for your user (recommended) or for all users
- Make sure your Advanced Installation Options look like this:



Launch the installer (cont'd 3) [Windows Only]

- After the installation type in the searchbar "Anaconda Powershell Prompt" and click on it to open the terminal
- Run the command `python --version` to check that the installation was successful. You should see the version of Python installed on your machine.
- Run the command `conda init powershell` to make conda available in the PowerShell.
- You should see something similar to this:



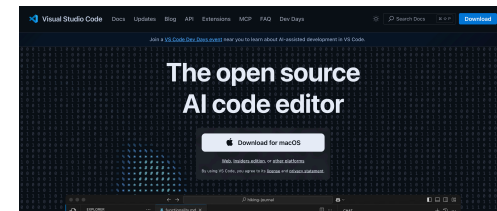
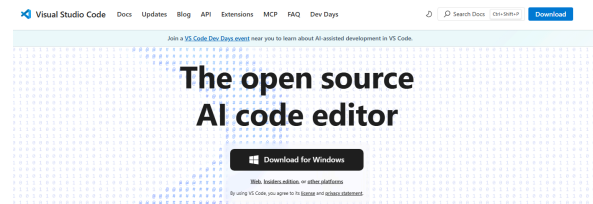
```
(base) PS C:\Users\vboxuser> python --version
Python 3.14.7
(base) PS C:\Users\vboxuser> conda init powershell
no change      C:\Users\vboxuser\miniconda3\Scripts\conda.exe
no change      C:\Users\vboxuser\miniconda3\Scripts\conda-script.py
no change      C:\Users\vboxuser\miniconda3\condabin\conda.bat
no change      C:\Users\vboxuser\miniconda3\Library\bin\conda.bat
no change      C:\Users\vboxuser\miniconda3\condabin\_conda_activate.bat
no change      C:\Users\vboxuser\miniconda3\condabin\rename_tmp.bat
no change      C:\Users\vboxuser\miniconda3\condabin\conda_auto_activate.bat
no change      C:\Users\vboxuser\miniconda3\condabin\conda_hook.bat
no change      C:\Users\vboxuser\miniconda3\Scripts\activate.bat
no change      C:\Users\vboxuser\miniconda3\condabin\activate.bat
no change      C:\Users\vboxuser\miniconda3\condabin\deactivate.bat
modified       C:\Users\vboxuser\miniconda3\Scripts\activate
modified       C:\Users\vboxuser\miniconda3\Scripts\deactivate
modified       C:\Users\vboxuser\miniconda3\etc\profile.d\conda.sh
modified       C:\Users\vboxuser\miniconda3\etc\fish\conf.d\conda.fish
no change      C:\Users\vboxuser\miniconda3\shell\condabin\Conda.ps1
modified       C:\Users\vboxuser\miniconda3\shell\condabin\conda-hook.ps1
no change      C:\Users\vboxuser\miniconda3\Lib\site-packages\xontrib\conda.xsh
modified       C:\Users\vboxuser\miniconda3\etc\profile.d\conda.csh
modified       C:\Users\vboxuser\Documents\WindowsPowerShell\profile.ps1

==> For changes to take effect, close and re-open your current shell. <==

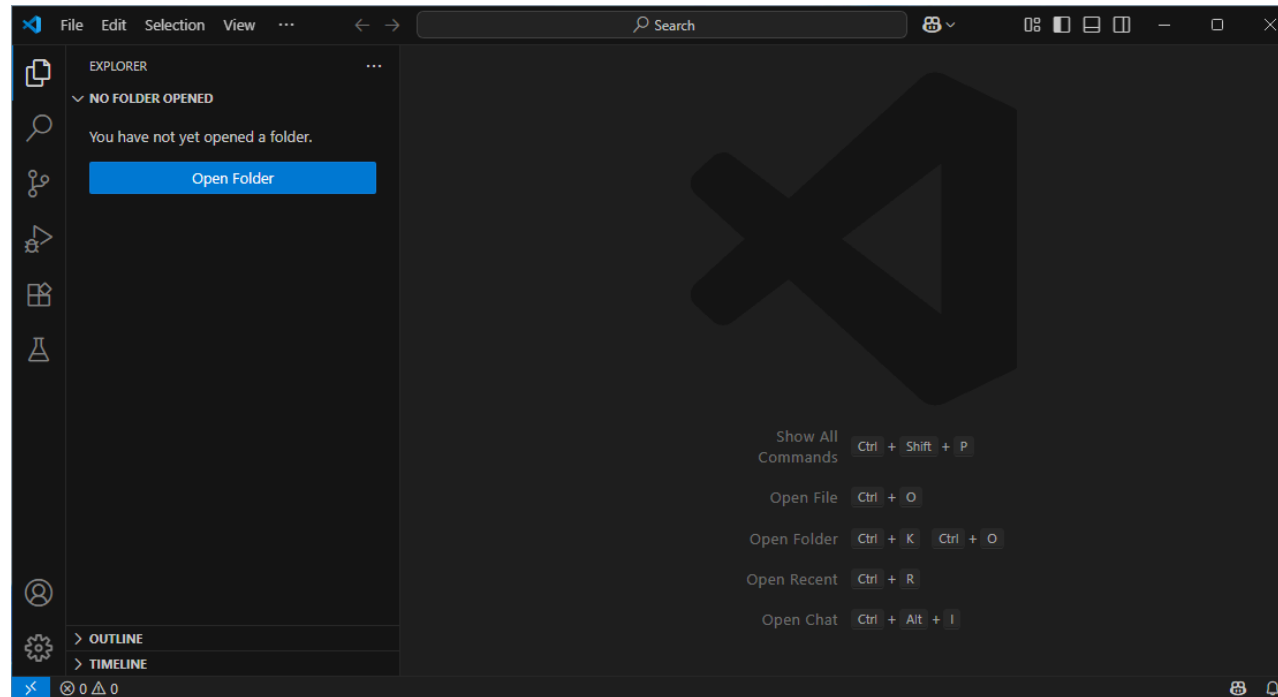
(base) PS C:\Users\vboxuser>
```

Install a (good) code editor: **Visual Studio Code**

1. Visit: <https://code.visualstudio.com>
2. Click on the button "Download for Windows" (if you have windows) or "Download for Mac" to start the download [check needed]
3. Install the VSCode and follow the instructions of the setup wizard
4. During installation, check the box "Add "Open with Code" action to Windows Explorer directory context menu." (windows only)

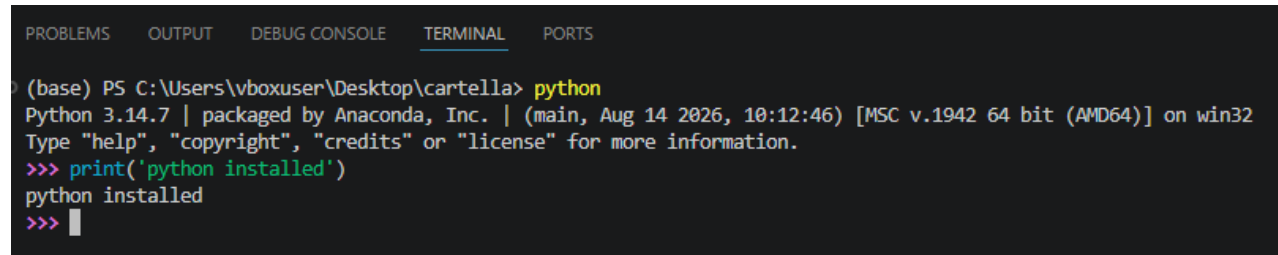


Start Visual Studio Code



Test python installation

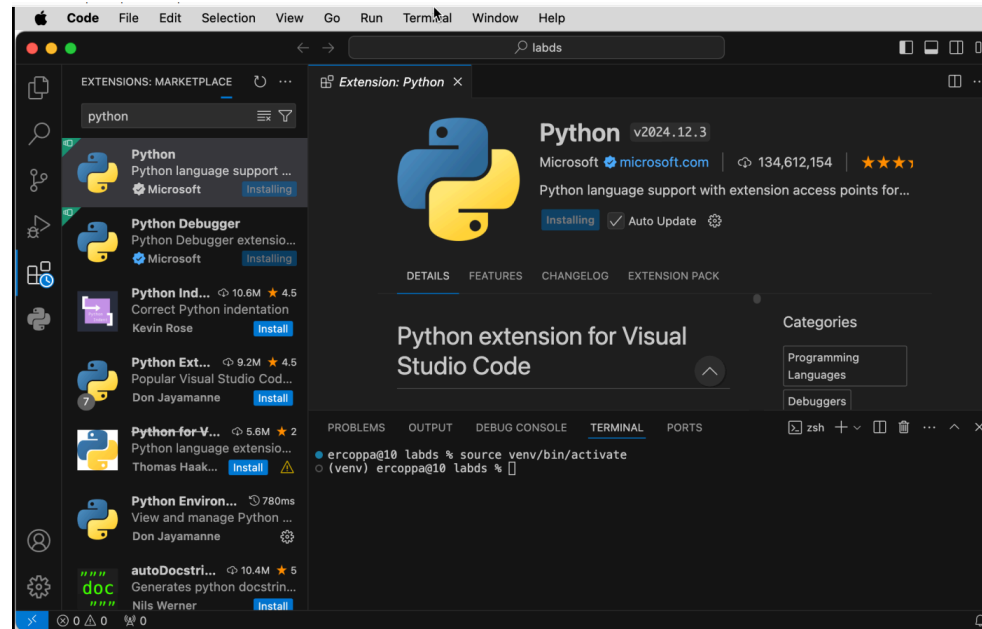
1. In VS Code start the terminal: ***Terminal > New Terminal***
2. Type python to open the interactive shell, a prompt input will appear with symbols >>>
3. Type the code `print('python installed!')` and hit enter, you will see the shell printing on screen "python installed!"



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
(base) PS C:\Users\vboxuser\Desktop\cartella> python
Python 3.14.7 | packaged by Anaconda, Inc. | (main, Aug 14 2026, 10:12:46) [MSC v.1942 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> print('python installed')
python installed
>>> 
```

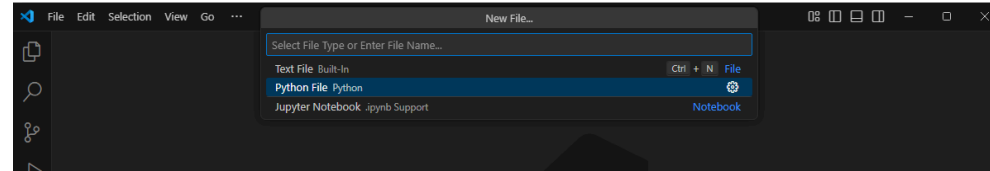
VS Code extension: Python integration

1. From the side bar, open the **Extension** tab
2. Search for **Python**
3. Install the extension

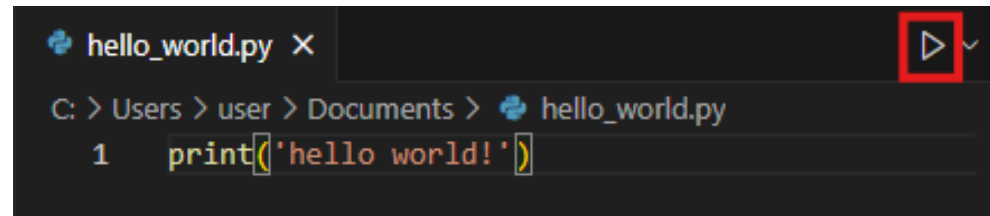


Create and run a python file in VS Code

1. In Visual Studio Code **File > New File**
2. Select **Python File**



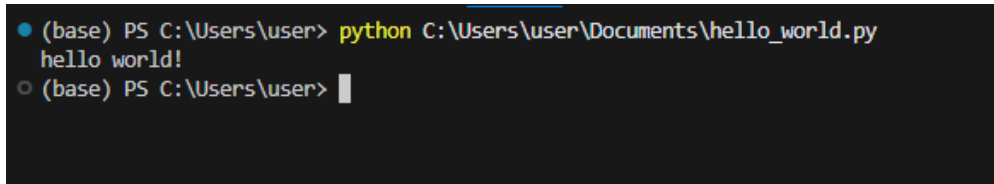
3. In the opened file type `print('hello world!')` and save the file
4. Click on the **play button** on the top right corner and the code will be executed in a new terminal



Create and run a python file in VS Code (cont'd)

When you hit play, VS Code is just running a command line in the terminal. That's another way to run python files:

1. In Visual Studio Code open the terminal **File > New Terminal**
2. Type `python <full-path-to-the-file>` and hit enter
(e.g. on Windows if you have a file named "hello_world.py" in the Documents folder)

A screenshot of a terminal window with a dark background. The first line shows a command prompt with the text "(base) PS C:\Users\user> python C:\Users\user\Documents\hello_world.py". The second line shows the output "hello world!". The third line shows the prompt "(base) PS C:\Users\user>" followed by a white cursor bar.

```
(base) PS C:\Users\user> python C:\Users\user\Documents\hello_world.py
hello world!
(base) PS C:\Users\user> 
```

VS Code extension: Jupyter

1. From the side bar, open the **Extension** tab
2. Search for **Jupyter**
3. Install the extension



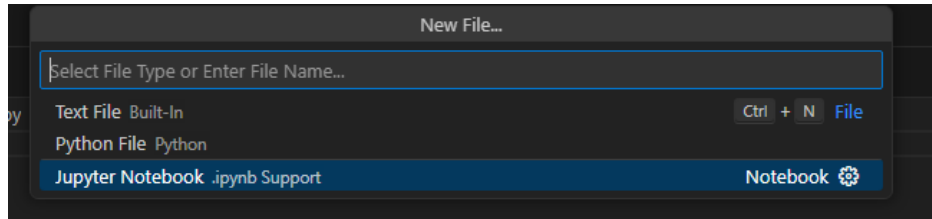
VS Code extension: Jupyter

In order for python notebooks to be run in VS Code, you need to install ipykernel.

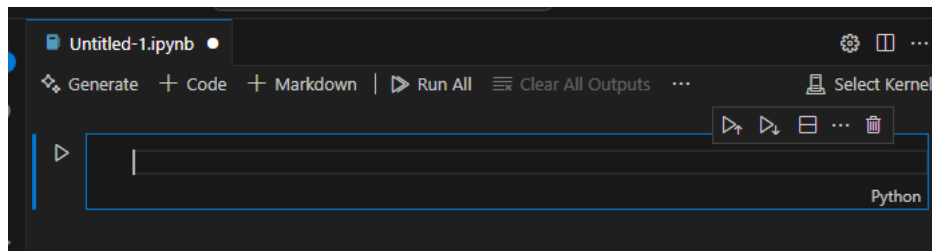
1. Open the terminal
2. type `pip install ipykernel`, hit enter and wait for the installation to finish

Create a new jupyter notebook

1. In Visual Studio Code **File** > **New File** tab
2. Select **Jupyter Notebook**

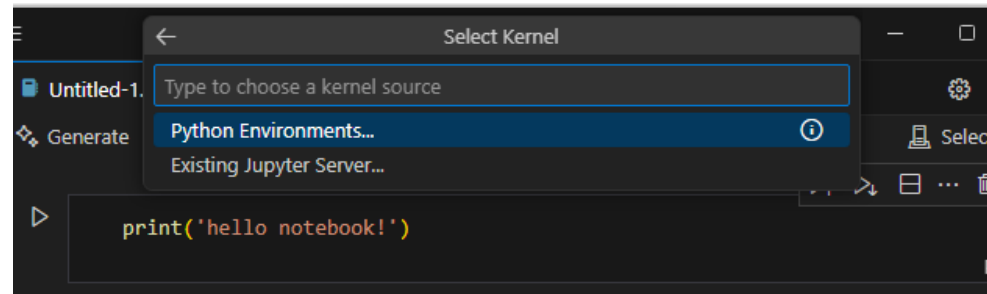


3. A new notebook will open



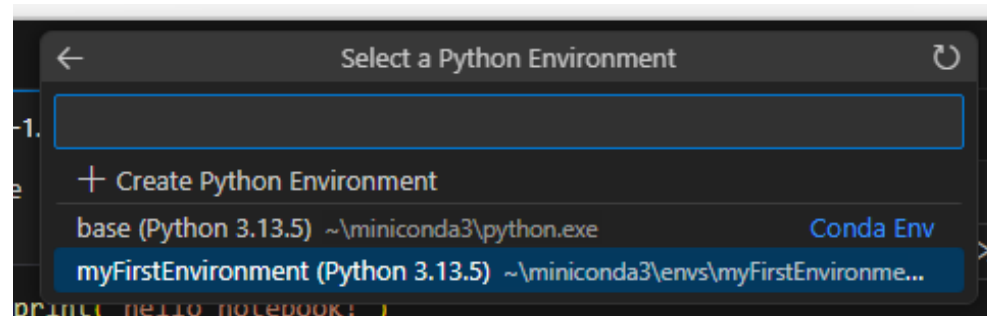
Running code in Jupyter notebook

1. In Visual Studio Code **File > New File > Jupyter Notebook**
2. When you create a new notebook, the new file contains a new code cell.
3. In the code cell type `print('hello world!')`
4. In the top right corner click on **Select Kernel > Python Environments...**

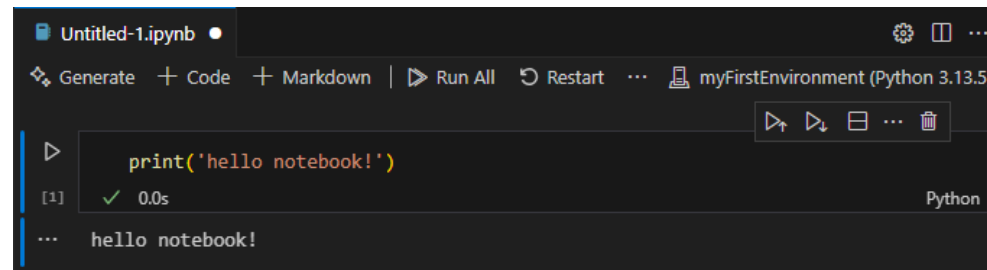


Running code in Jupyter notebook (cont'd)

5. Choose the python environment you've created for your project



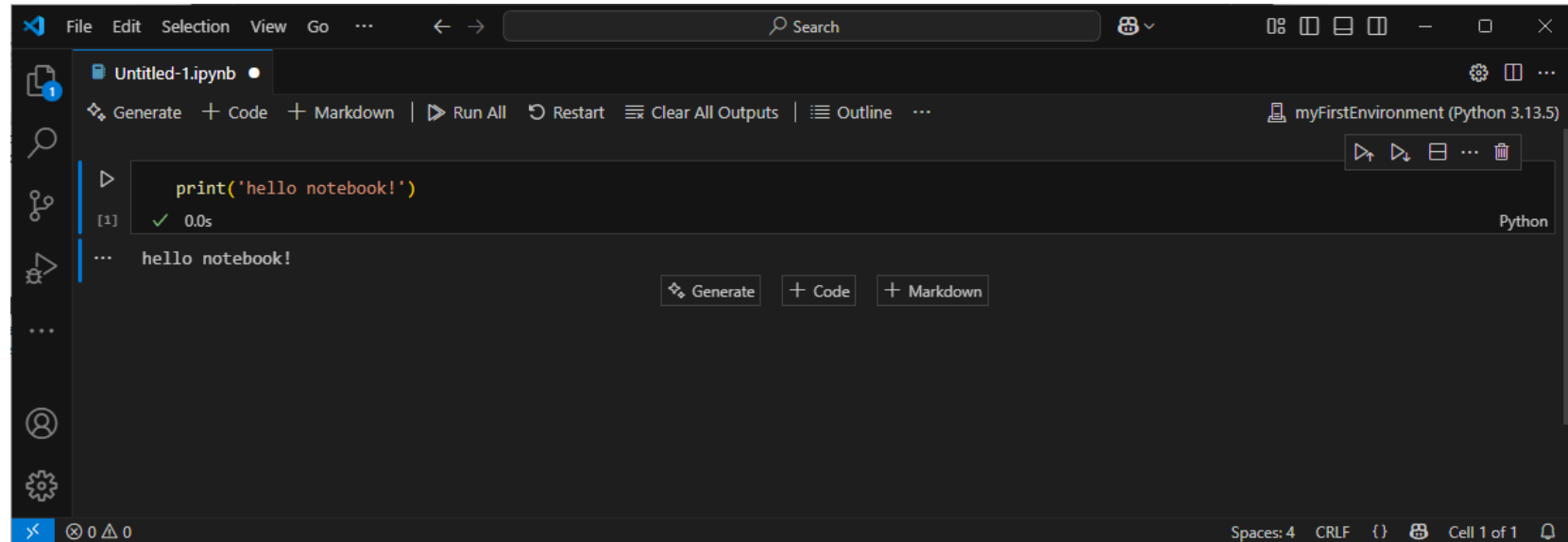
6. Click the play button on the left side of the cell. The code will run and the result will appear under the code cell



Working with jupyter

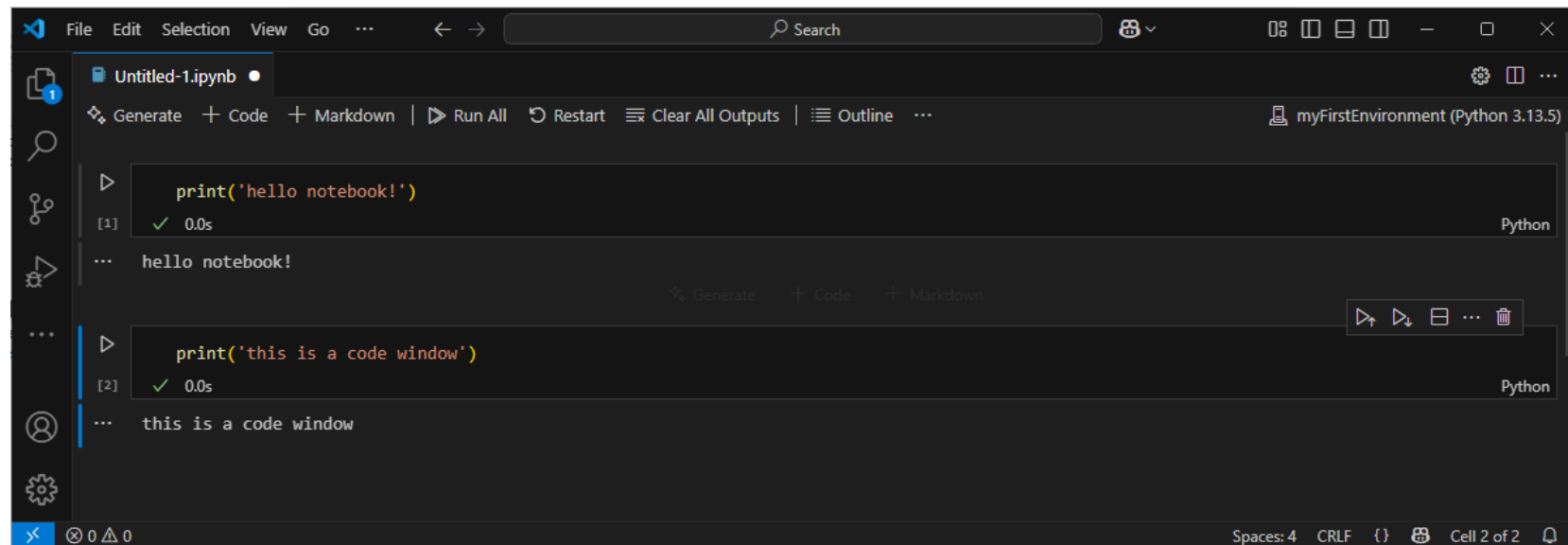
In a Jupyter notebook you can add two different types of cell

1. Code - to write code to run
2. Markdown - to add text and format it



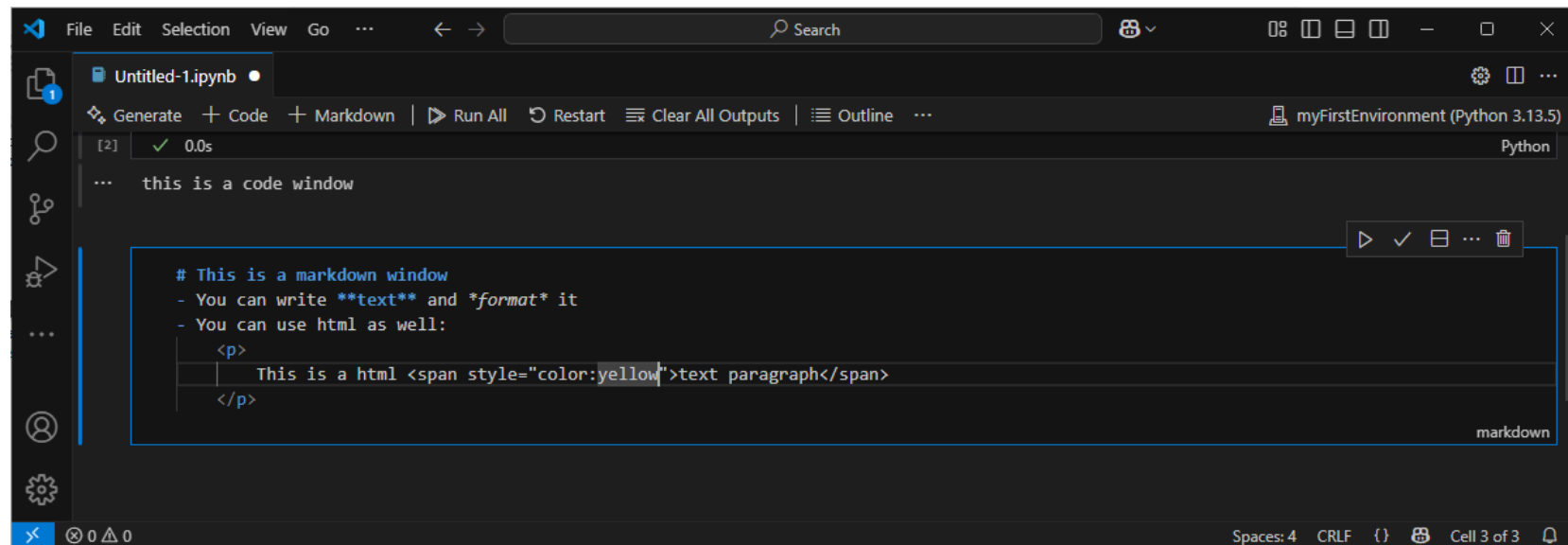
Working with jupyter - Code cell

1. Click on the "+ Code" button
2. Write python code
3. Run it (play button or *shift + enter*)



Working with jupyter - Markdown cell

1. Click on the "+ Markdown" button
2. Write text and format it with markdown symbols
3. You can also use html



The screenshot shows a Jupyter Notebook window titled "Untitled-1.ipynb". The interface includes a menu bar (File, Edit, Selection, View, Go), a search bar, and a toolbar with buttons for Generate, Code, Markdown, Run All, Restart, Clear All Outputs, and Outline. The environment is set to "myFirstEnvironment (Python 3.13.5)".

The notebook contains two cells:

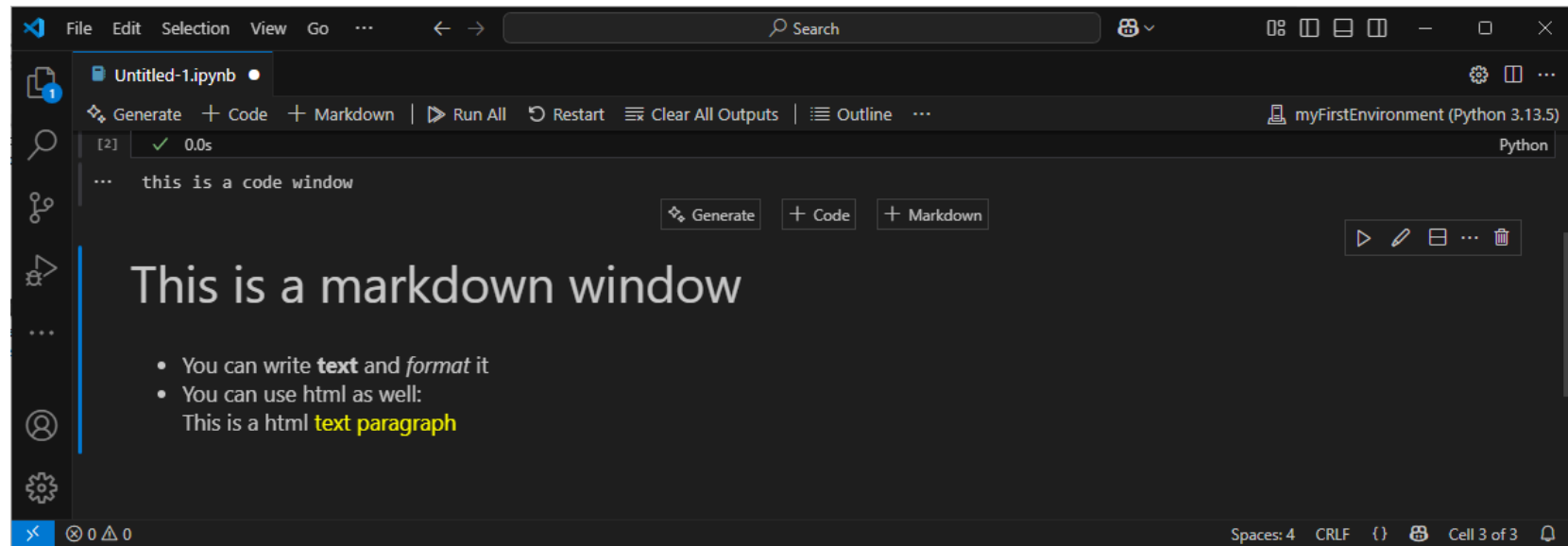
- A code cell (labeled [2]) with the text "this is a code window".
- A markdown cell (labeled markdown) containing the following text:

```
# This is a markdown window
- You can write text and format it
- You can use html as well:
  <p>
    This is a html <span style="color:yellow">text paragraph</span>
  </p>
```

The status bar at the bottom indicates "Spaces: 4", "CRLF", and "Cell 3 of 3".

Working with jupyter - Markdown cell (cont'd)

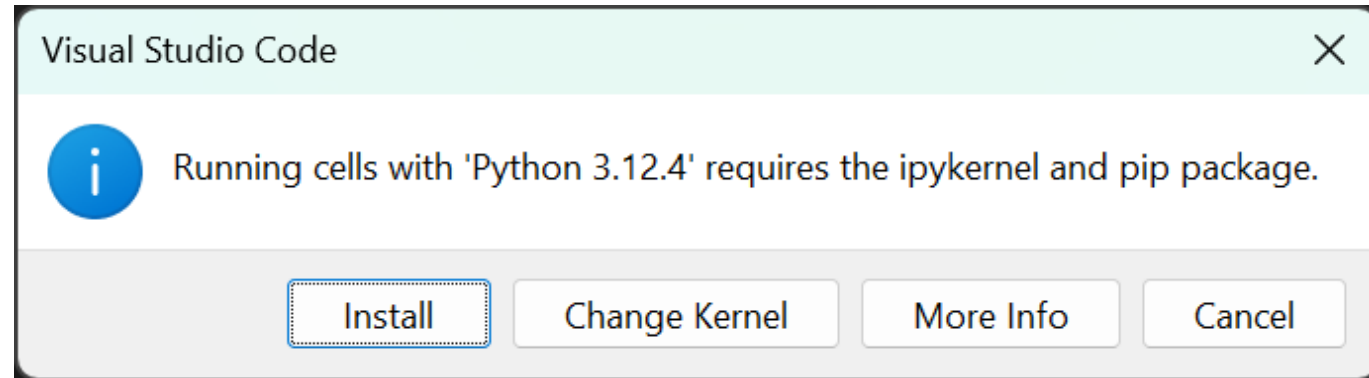
4. Confirm it (Check symbol on the top right or *shift + enter*) to see the result



Troubleshooting

Troubleshooting first ipynb run

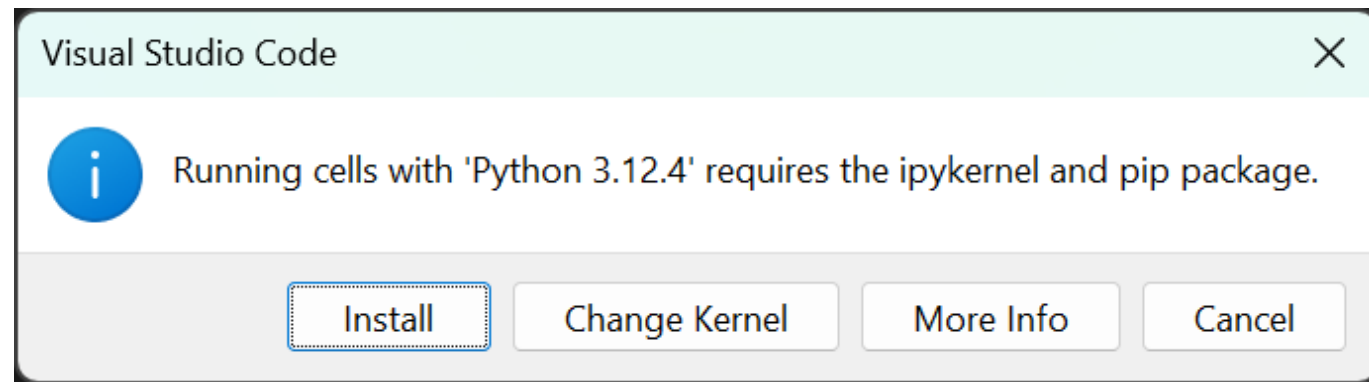
If, when when running you first notebook inside Visual Studio Code, you get this this window:



Click on *Install* to install the missing ipykernel

Troubleshooting first ipynb run (cont'd)

If doing the installation, this error appears at the bottom right corner of VS Code



And this error under your code cell

```
Running cells with 'Python 3.12.4' requires the ipykernel package.  
Create a Python Environment with the required packages.  
Or install 'ipykernel' using the command: 'c:/Python312/python.exe -m pip install ipykernel -U --user --force-reinstall'
```

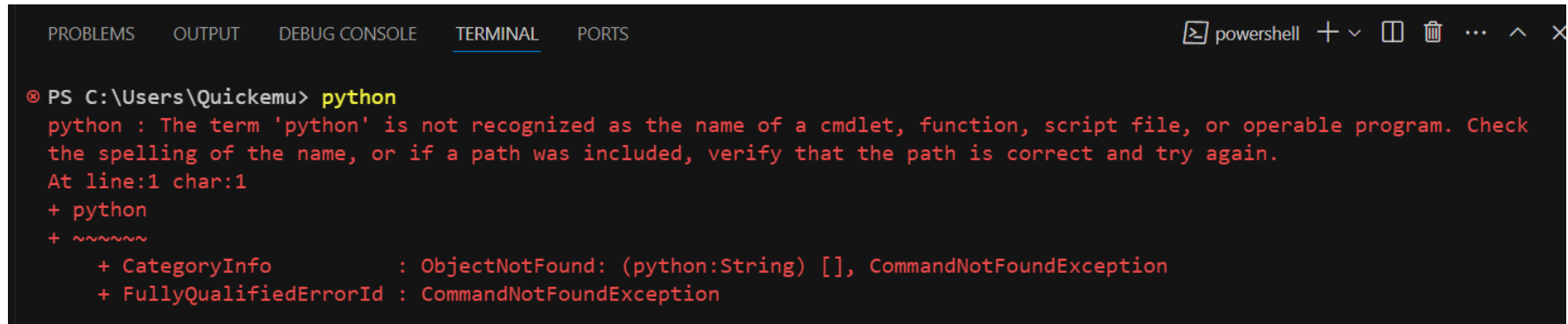
Copy the command and paste it in the terminal `c:/Python312/python.exe -m pip install ipykernel -U --user --force-reinstall`

Troubleshooting on Windows

Troubleshooting on Windows:

`python` is not recognized

If, when when running python in the terminal of VSCode, you get the following error:

A screenshot of a VS Code terminal window. The terminal title bar shows 'powershell' and standard window controls. The terminal content shows a PowerShell prompt 'PS C:\Users\Quickemu>' followed by the command 'python' in yellow. The output is in red text: 'python : The term 'python' is not recognized as the name of a cmdlet, function, script file, or operable program. Check the spelling of the name, or if a path was included, verify that the path is correct and try again. At line:1 char:1', followed by a red prompt '+ python' and another red prompt '+ ~~~~~'. Below this, two lines of red text provide error details: '+ CategoryInfo : ObjectNotFound: (python:String) [], CommandNotFoundException' and '+ FullyQualifiedErrorId : CommandNotFoundException'.

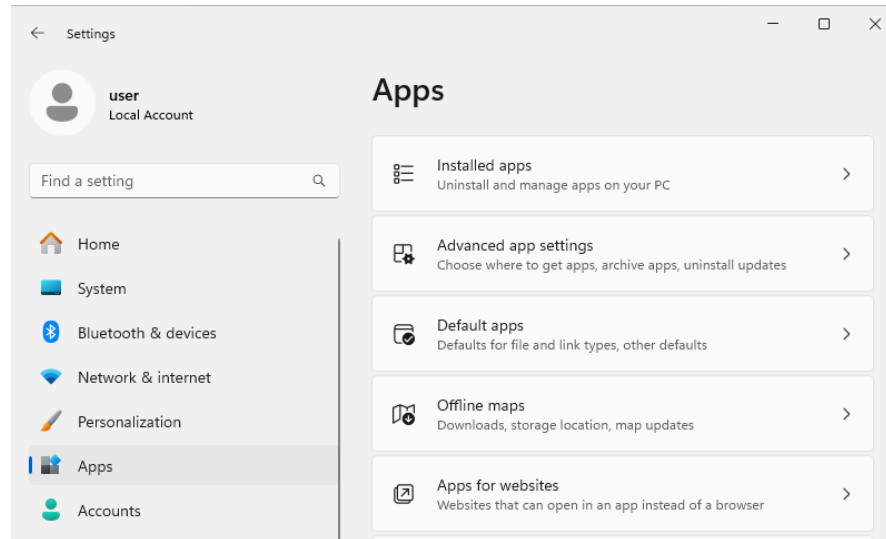
```
PS C:\Users\Quickemu> python
python : The term 'python' is not recognized as the name of a cmdlet, function, script file, or operable program. Check
the spelling of the name, or if a path was included, verify that the path is correct and try again.
At line:1 char:1
+ python
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (python:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException
```

Then, you have to:

Troubleshooting on Windows: python is not recognized (cont'd)

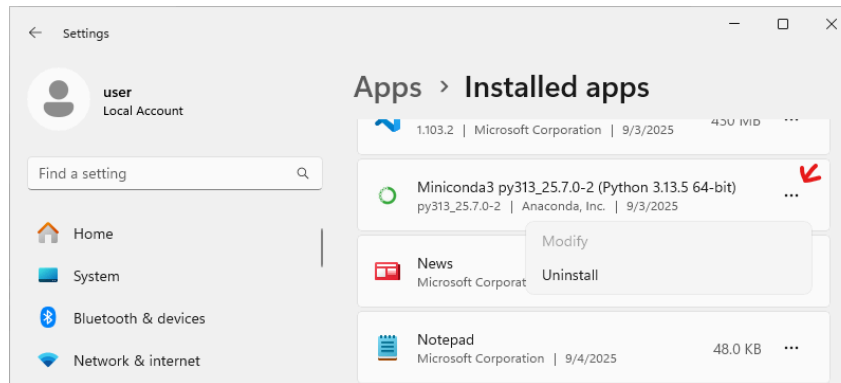
STEP 1 Uninstall Miniconda:

1. Click on **Start > Settings**
2. Select **Apps** from the left menu and then click **Installed apps**



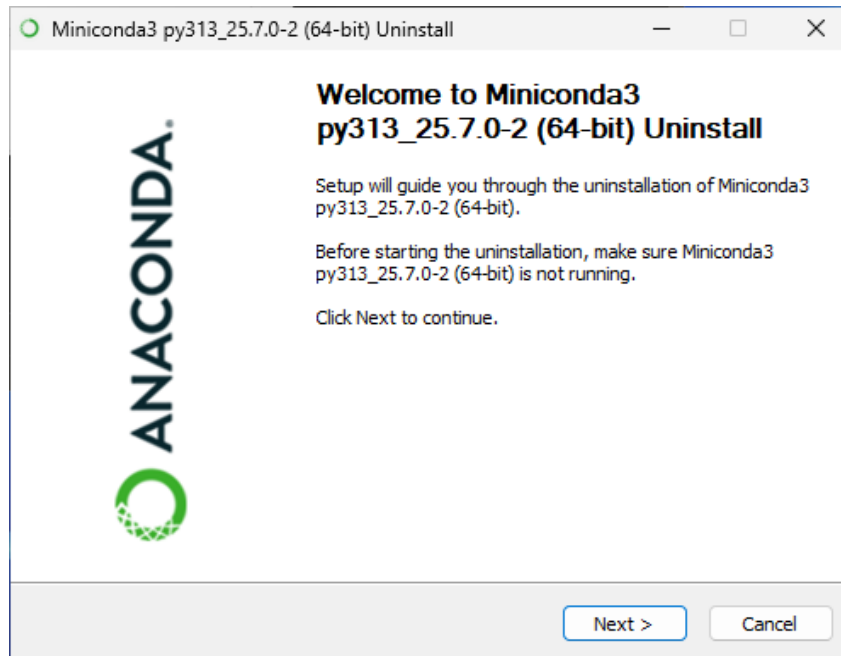
Troubleshooting on Windows: python is not recognized (cont'd - 2)

3. Look for Miniconda, click on **...** > **Uninstall**



Troubleshooting on Windows: python is not recognized (cont'd - 3)

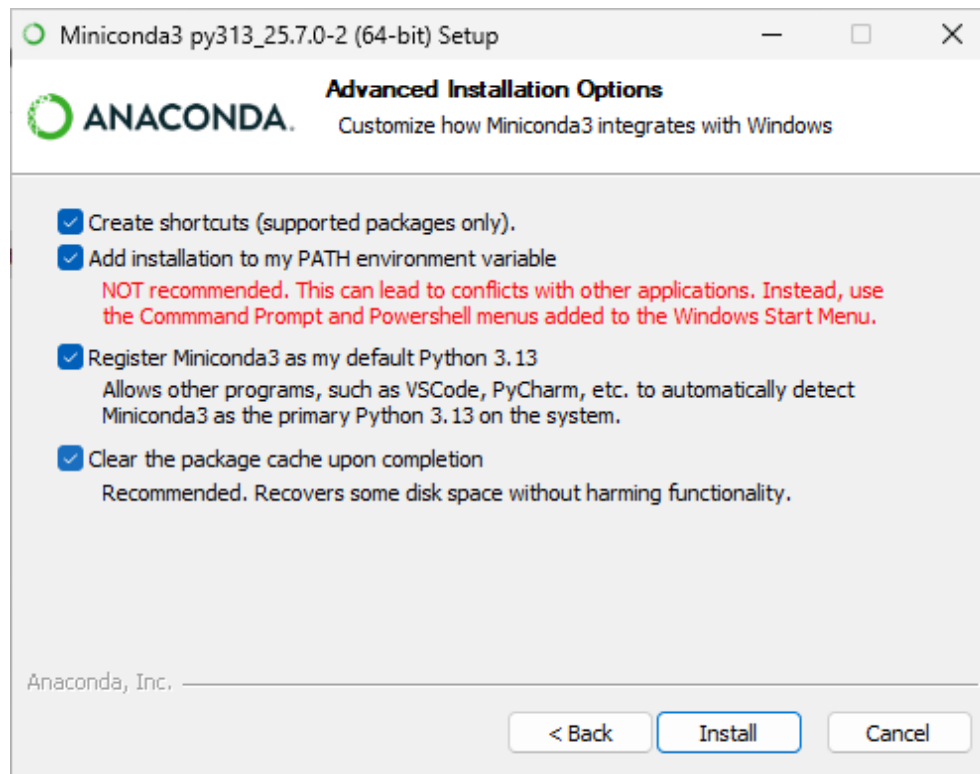
4. Run the conda uninstaller



Troubleshooting on Windows: python is not recognized (cont'd - 4)

STEP 2 Reinstall miniconda but add it to PATH this time:

1. Start the miniconda installer
2. Click next and accept license terms
3. In the *Advanced Installation Options* view, check "Add installation to my PATH environment variable", ignore warnings. Then continue the setup.



Cloud Setup

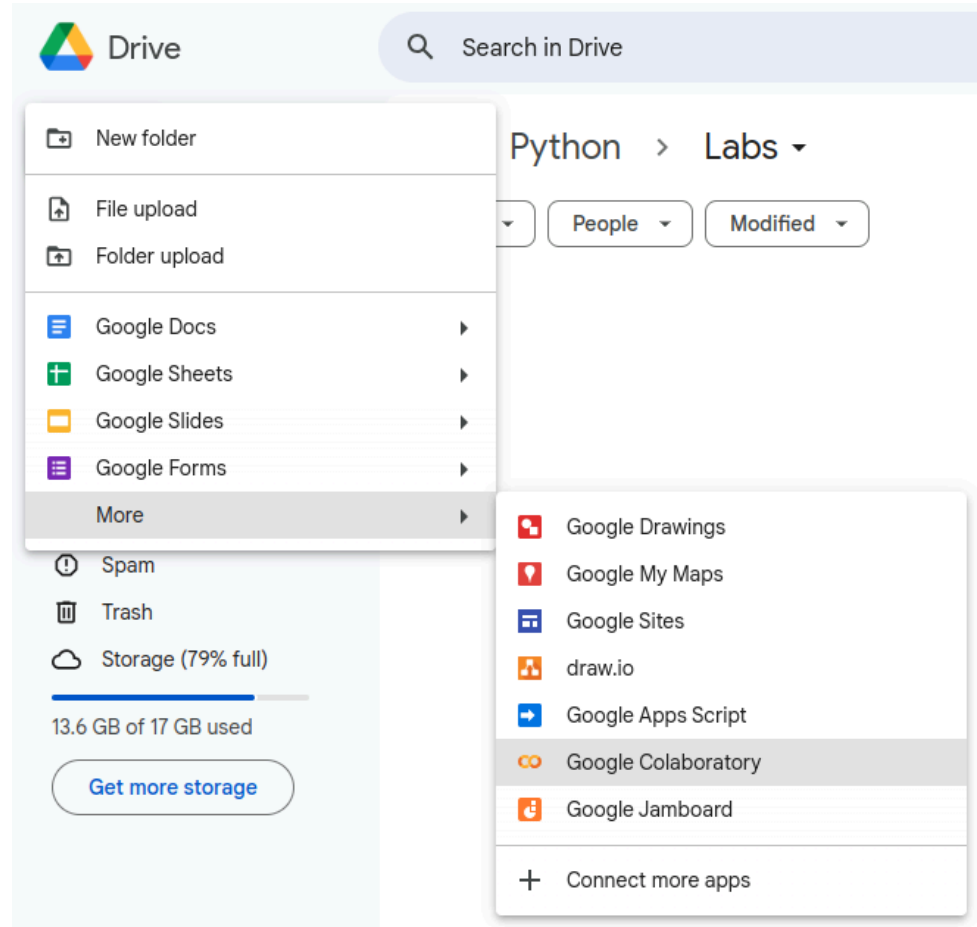
Google Colab, i.e., the lazy approach



- Platform from Google
- **Google Account is required**
- Free of charge to use for the basic plan
- Write code in the browser (no extra is needed)
- Code executed on Google cloud servers
- Low/medium performance, privacy issues
- **No need to install anything**

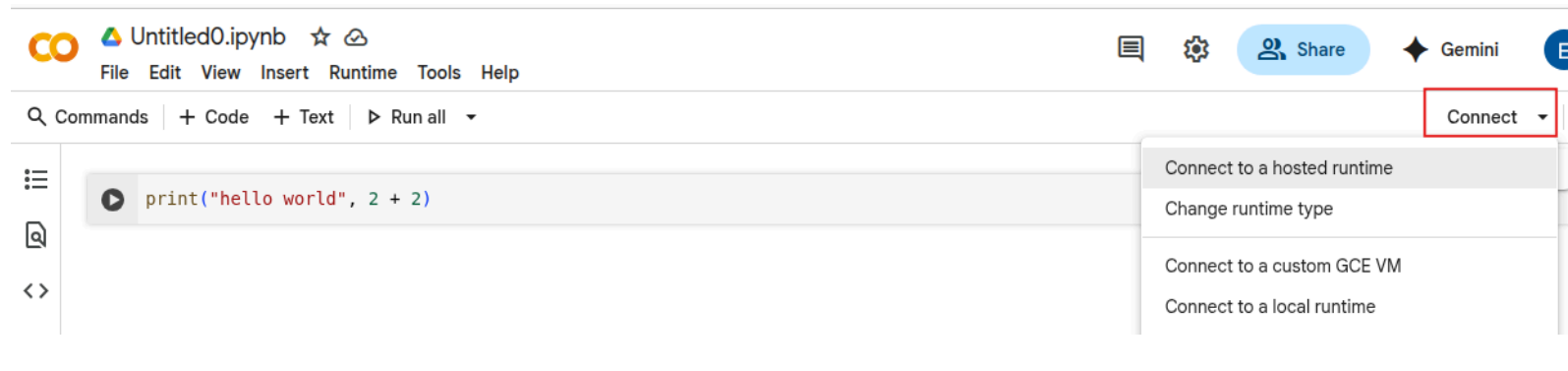
Google Colab, i.e., the lazy approach (cont'd)

- From **Google Drive**



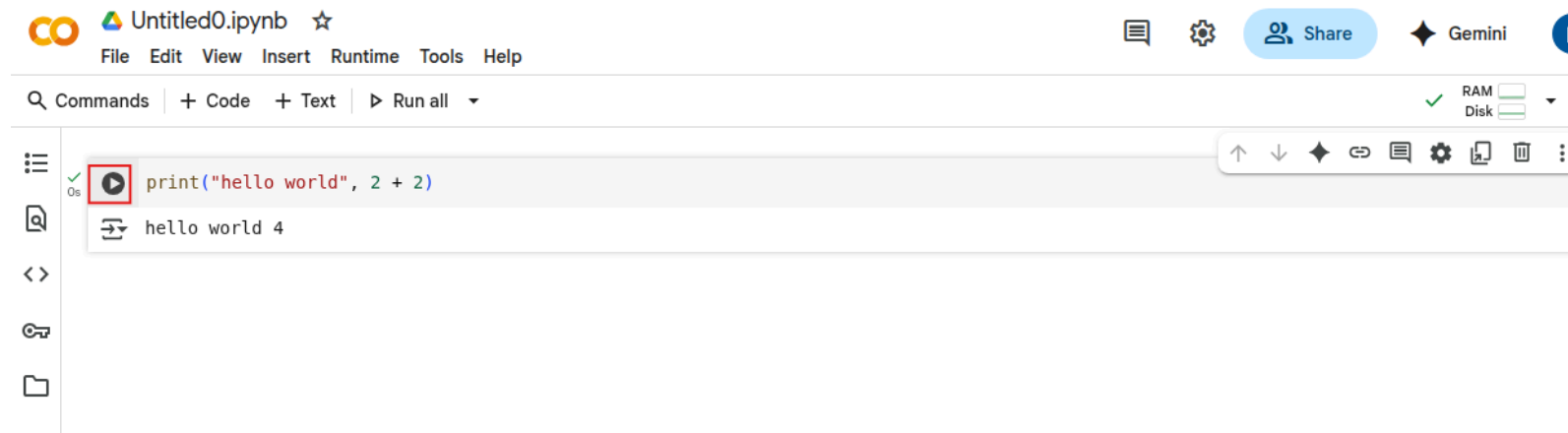
New Colab notebook

- A new notebook will open in colab
- Use as usual
- Before running you must connect to a runtime



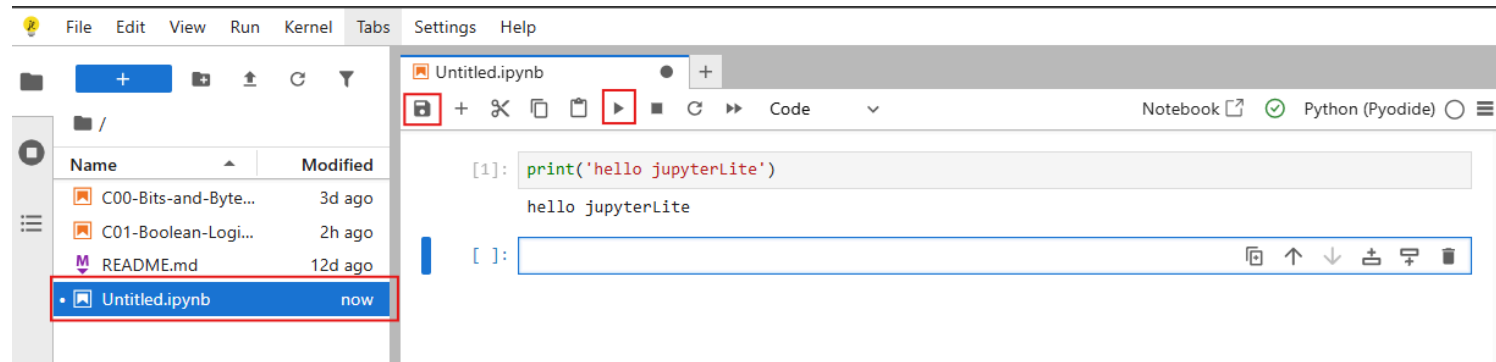
New Colab notebook (cont'd)

- Once connected, write code and run code in cells pressing the play button on the left side of the cell



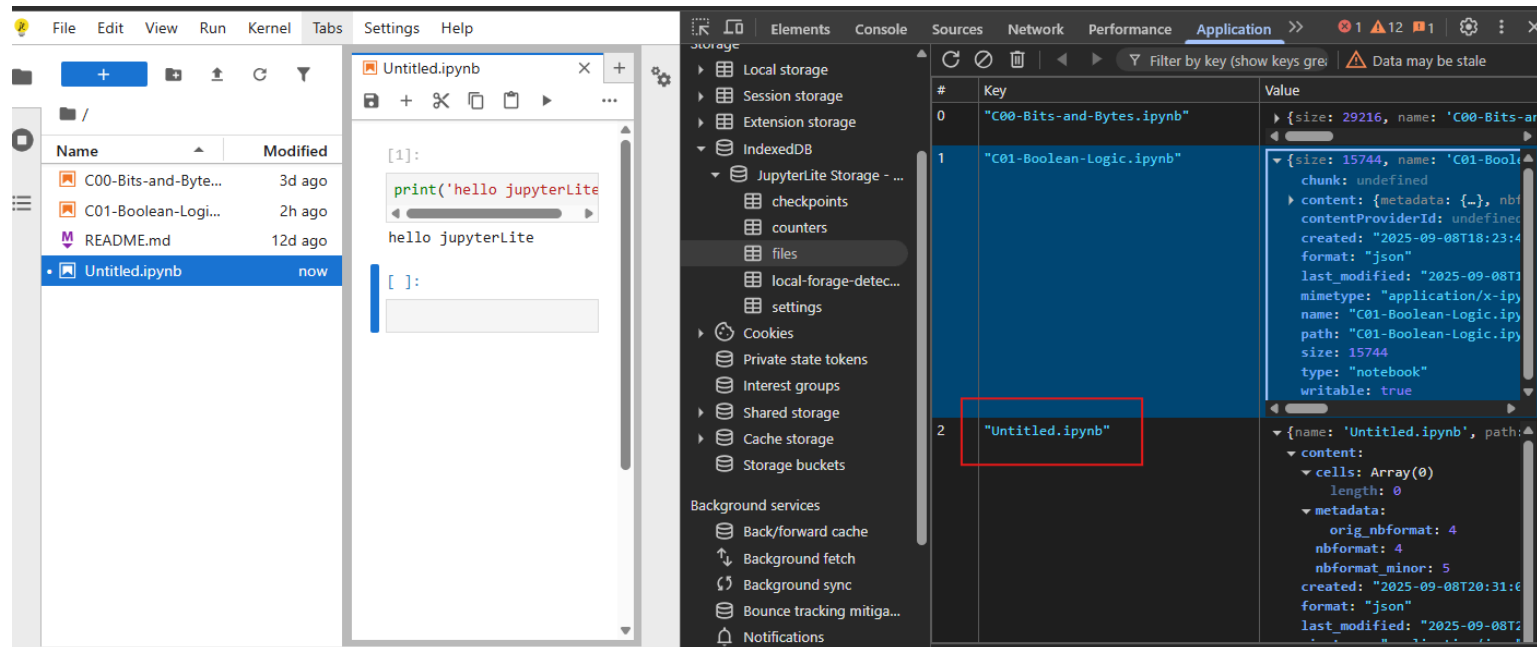
JupyterLite

- <https://ercoppa.github.io/jupyterlite/>
- **It does not require an account**
- Add code and cells
- Run with play button (or *shift + enter*)
- Save your file



JupyterLite (cont'd)

- Your files are saved in the browser's cache
- **You will lose them when you'll delete your "cookies and other site data"**



Environments

Environment: local setup with VSCode

Use VSCode:

- Quickly write and execute python files (.py)
- Quick access to the local terminal
- Write and execute notebooks (.ipynb). Some advanced features of JupyterLab (that we do not need) may have limited support.
- Good editor with several extensions (integrations with third-party services)
- Everything runs on your machine

This is the environment that we suggest you to use.

Environment: cloud setup with Colab

Use Google Colab:

- You can work from any computer. Even your tablet (although, it is a bit inconvenient!)
- Write and execute on the cloud notebooks (.pynb). Some advanced features of JupyterLab (that we do not need) may have limited support.
- Basic editor
- Everything must be saved on Google Drive
- **Not adequate in future courses where you need a bit of computational power** (assuming you do not want to pay Google).

We suggest to use Colab only if you have issues with your local setup

Environment: cloud setup with JupyterLite

Use JupyterLite:

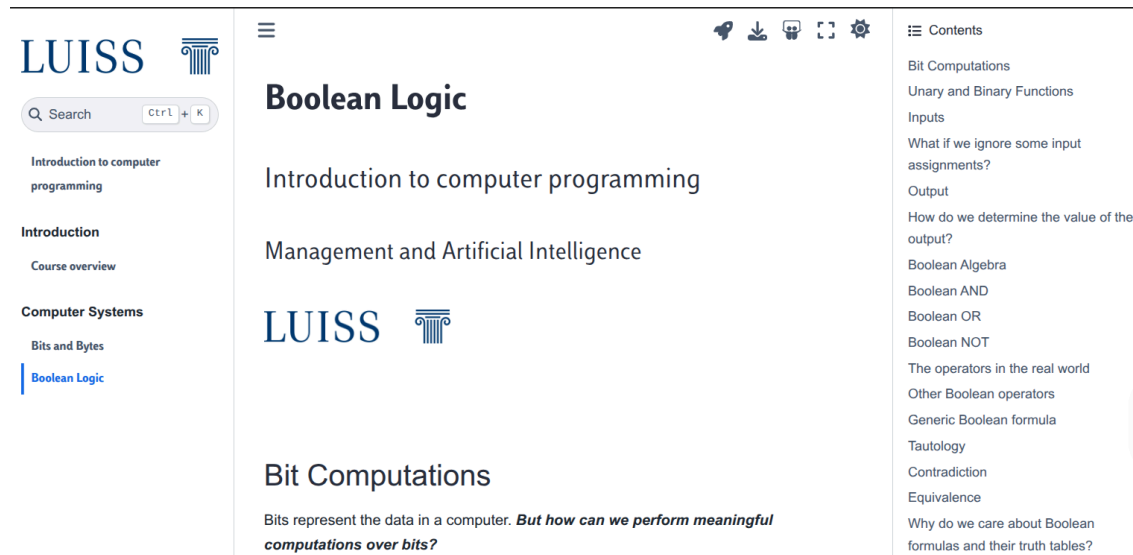
- You can work from any computer. Even your tablet (although, it is a bit inconvenient!)
- Write and execute notebooks (.ipynb).
- Everything is saved in the browser's local database
- Everything runs actually on your machine

We suggest to use JupyterLite only when you cannot use VSCode and do not want use Colab

Using the book

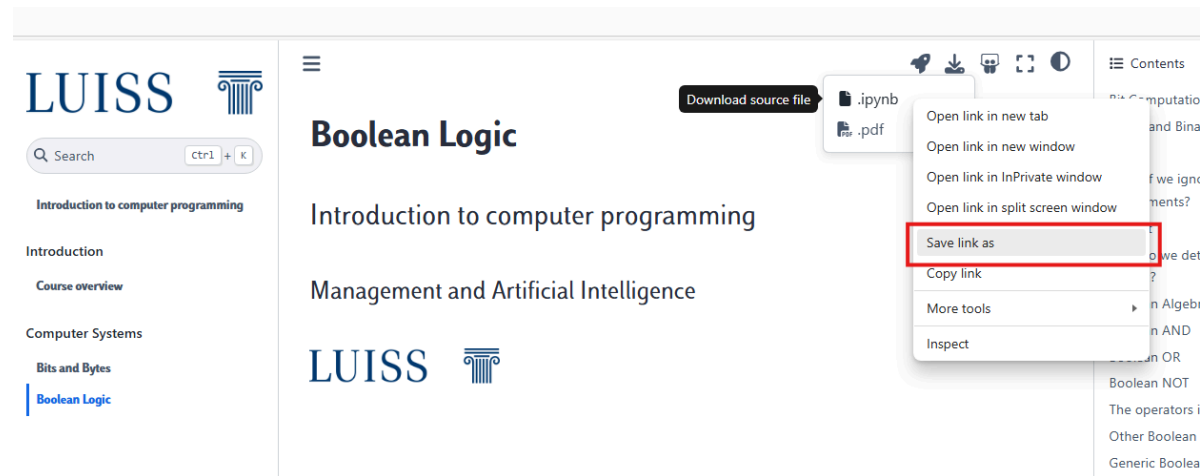
Accessing the material on the book

1. Visit <https://introcp.github.io/> to access the book
2. Use the left menu to get the lesson content



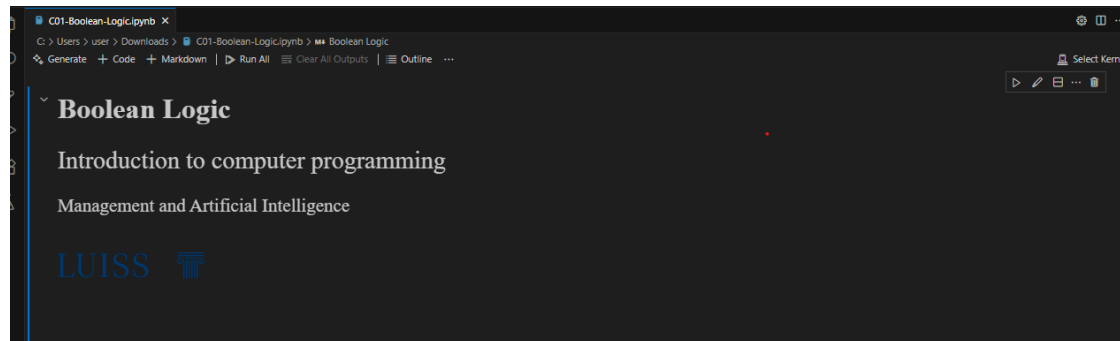
Download notebooks from the book

1. Click on the  icon
2. Right click on **.ipynb** > **Save link as** to save the file on your pc



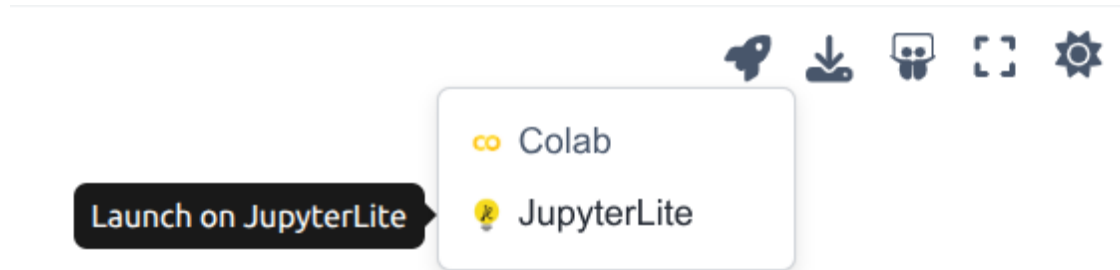
Opening notebooks from the book in VSCode

3. Open the .ipynb in VS Code
4. Add your notes and code snippets



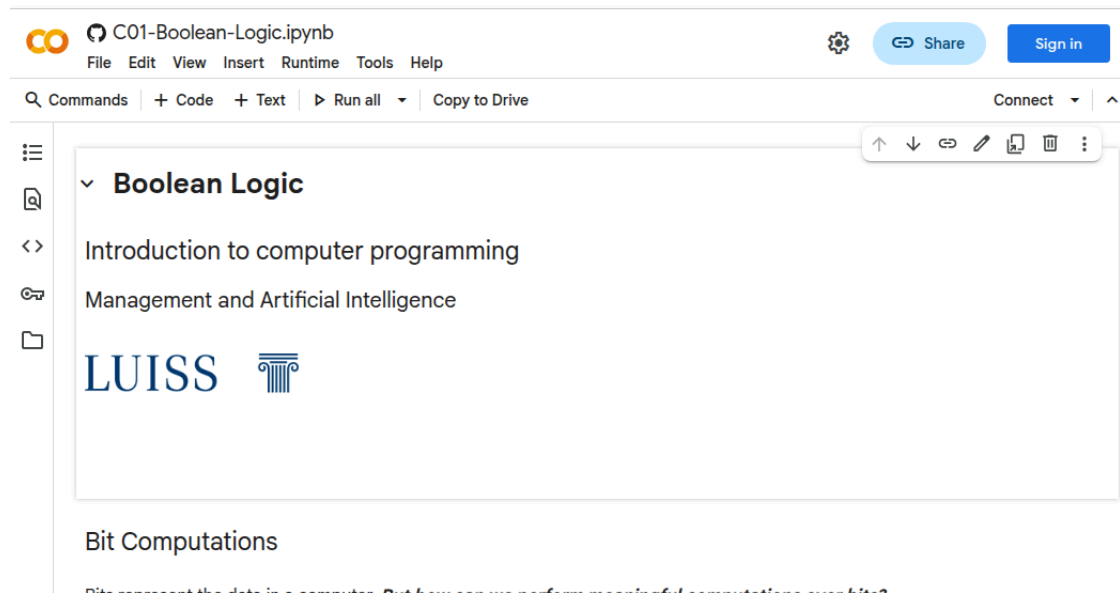
Opening notebooks from the book: Colab or JupyterLite

1. Click on the  icon
2. Click on Colab/JupyterLite



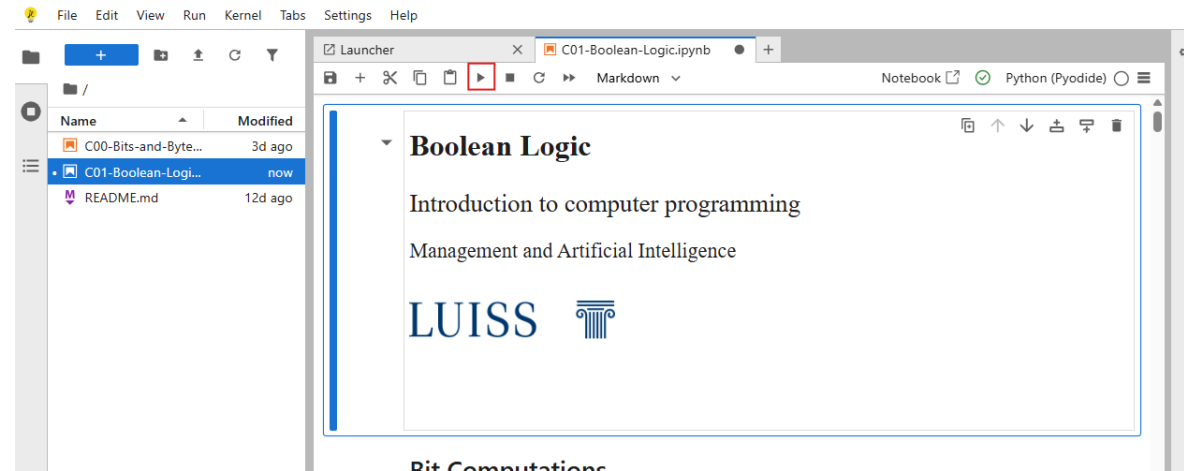
Importing notebooks from the book: Colab

- Import a notebook in colab
- Add your notes or code snippets



Importing notebooks from the book: JupyterLite

- Import the notebook in JupyterLite
- Add your notes or code snippets
- Run code snippets by clicking on the play button (on the top of the notebook or on the left of the code cell)



Importing notebooks from the book: why you should

1. Add your personal notes and thoughts notes to theory slides during the lectures!
2. Work on the Python exercises and run examples without further efforts
3. Active learning (and learning by doing) is a proven way to enhance learning experience and results

Python Virtual Environments

This will be useful for future courses!

Where to install Python packages from the community

We can install Python packages from the community following two strategies:

- **System-wide**: this may create conflicts between different projects using Python.
 - pros: you need to install a package only once for all projects
 - cons: one project may break due to a package installed/updated for another project

It is now the deprecated approach.

- **Project-wide**: each project has its own **Virtual Environment**.
 - pros: no conflicts across packages of different projects
 - cons: a package must be installed again for each project

This is nowadays the suggested approach and **we will follow it.**

Create a new virtual environment [Only Windows]

1. In VS Code, start the terminal: ***Terminal > New Terminal***
2. Only for Windows: `conda init powershell`
Then close the terminal



Create a new virtual environment (cont'd)

1. In VS Code, start the terminal: ***Terminal > New Terminal***
2. Type in the terminal: `conda create --name <my-env>`
Where <my-env> is the name of your virtual environment
3. Conda will ask to proceed Proceed ([y]/n)?
Type y and hit enter to confirm

NOTE: The first time you create an environment, conda will ask to accept the terms of agreement ****before asking to proceed****

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
PS C:\Users\user\Documents\intro-python-25\MyEnv> conda create --name myFirstEnvironment
Do you accept the Terms of Service (ToS) for https://repo.anaconda.com/pkg/main? [(a)ccept/(r)eject/(v)iew]: a
Do you accept the Terms of Service (ToS) for https://repo.anaconda.com/pkg/r? [(a)ccept/(r)eject/(v)iew]: a
Do you accept the Terms of Service (ToS) for https://repo.anaconda.com/pkg/msys2? [(a)ccept/(r)eject/(v)iew]: a
3 channel Terms of Service accepted
```

Print the environment list

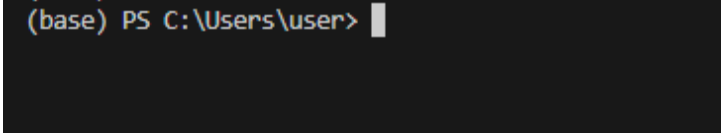
- To verify that the new environment was created correctly, we'll print the list of all the existing environments:
`conda env list`
- The newly created environment name should appear in the list

```
PS C:\Users\user\Documents\intro-python-25\MyEnv> conda env list

# conda environments:
#
base                C:\Users\user\miniconda3
myFirstEnvironment  C:\Users\user\miniconda3\envs\myFirstEnvironment
```

How to use environments while working

- There is a default **base** environment.

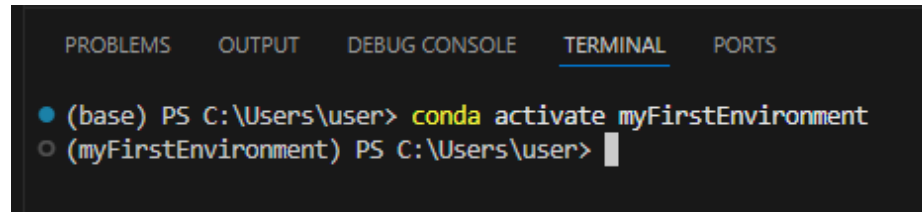
A screenshot of a terminal window with a black background. The text "(base) PS C:\Users\user>" is displayed in a light blue font, followed by a white cursor bar.

```
(base) PS C:\Users\user>
```

- When starting a new project, we will create a new environment
- Then our workflow will be like:
 1. Before starting to edit a project: **activate** the environment created for that project: `conda activate <my-env>`
 2. While working on the project: install any **python package** needed `conda install <package-name>`
 3. When finished: **deactivate** the environment (to avoid conflicts between projects) `conda deactivate`

Activate you environment

1. Type in the terminal: `conda activate <my-env>`
Where `<my-env>` is the name of your virtual environment
2. The name of the environment will replace of "(base)" at the beginning of the command line, to confirm the environment is active.



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

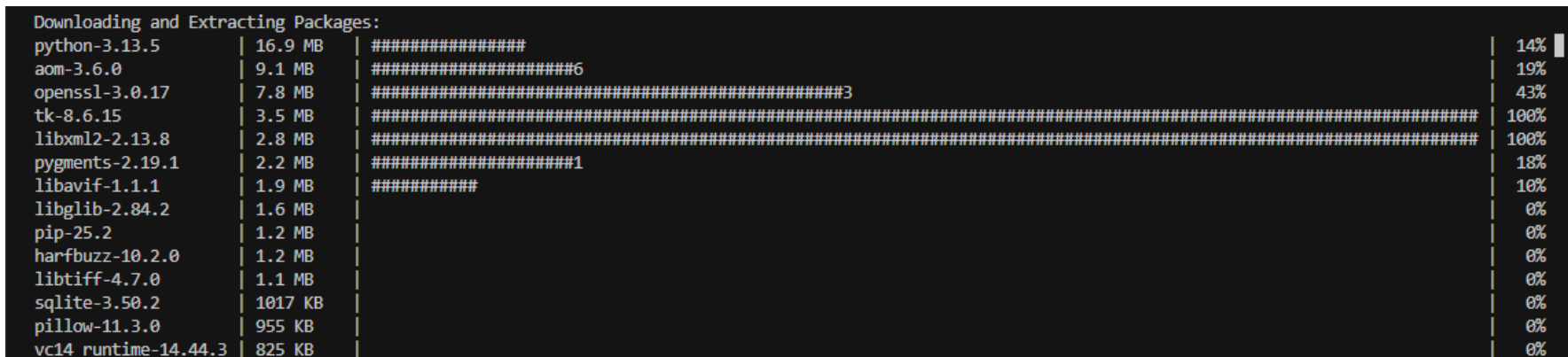
● (base) PS C:\Users\user> conda activate myFirstEnvironment
○ (myFirstEnvironment) PS C:\Users\user> 
```


Installation of Python packages in the virtual environment

We will use some extra Python bits from the community, including:
numpy, matplotlib, seaborn, scikit-learn, and jupyterlab.

From VSCode:

1. Start a new terminal using the menu: ***Terminal > New Terminal***
2. `conda install numpy matplotlib seaborn scikit-learn jupyterlab`
3. When prompted to Proceed ([y]/n)? type y and hit enter. The package download will start



Downloading and Extracting Packages:			
python-3.13.5	16.9 MB	#####	14%
aom-3.6.0	9.1 MB	#####6	19%
openssl-3.0.17	7.8 MB	#####3	43%
tk-8.6.15	3.5 MB	#####	100%
libxml2-2.13.8	2.8 MB	#####	100%
pygments-2.19.1	2.2 MB	#####1	18%
libavif-1.1.1	1.9 MB	#####	10%
libglib-2.84.2	1.6 MB		0%
pip-25.2	1.2 MB		0%
harfbuzz-10.2.0	1.2 MB		0%
libtiff-4.7.0	1.1 MB		0%
sqlite-3.50.2	1017 KB		0%
pillow-11.3.0	955 KB		0%
vc14_runtime-14.44.3	825 KB		0%

